

CS677-Spring2025: Machine Learning

Assignment#2 - TOTAL POINTS: 100

DUE DATE: 03/27/2025 (Mar 27th)

(This is a Team assignment)

S.No.	Questions	Points	Self-Assessment										
1	Explain the example code step by step for the given reference links. For each python notebook, you must write an explanation of about 75-100 words: A. https://scikit-learn.org/stable/auto_examples/linear_model/plot_ols.html#linear-regression-example B. https://scikit-learn.org/stable/auto_examples/linear_model/plot_iris_logistic.html#logistic-regression-3-class-classifier C. https://scikit-learn.org/stable/auto_examples/model_selection/plot_underfitting_overfitting.html	25											
2	Navigate to this dataset: https://data.cityofnewyork.us/Education/Prelim-Average-Class-Size-School-MS-HS/d7b6-byut/data_preview What supervised learning problem statements are possible with the above dataset? Frame one predictive statement each for Classification and Regression tasks(or whichever you think is possible for the dataset).	25											
3	Write a python code to solve simple (Univariate) Linear Regression and predict house prices based on the size of the house using the hypothesis $h\theta(x) = \theta_0 + \theta_1x$? Plot best fit line. (Do not use any predefined functions) <table><tr><th>Size in feet² (x)</th><th>Price (\$) in 1000's (y)</th></tr><tr><td>2083</td><td>445</td></tr><tr><td>1237</td><td>304</td></tr><tr><td>1015</td><td>223</td></tr><tr><td>635</td><td>166</td></tr></table>	Size in feet ² (x)	Price (\$) in 1000's (y)	2083	445	1237	304	1015	223	635	166	25	
Size in feet ² (x)	Price (\$) in 1000's (y)												
2083	445												
1237	304												
1015	223												
635	166												

4	Write a python code to calculate loss function using empirical risk function (Mean Squared Error) $L_s(h) = 1/m \sum_{i=1}^m h((x^i) - y^i)^2$ (Do not use any predefined functions) <table><tr><th>Actual Value</th><th>Predicted value</th></tr><tr><td>19</td><td>20</td></tr><tr><td>26</td><td>30</td></tr><tr><td>30</td><td>23</td></tr><tr><td>45</td><td>36</td></tr></table>	Actual Value	Predicted value	19	20	26	30	30	23	45	36	25	
Actual Value	Predicted value												
19	20												
26	30												
30	23												
45	36												

Submission Instructions:

- Submit a Word document with a detailed explanation for each link provided for Questions 1 and 2.
- You may not be graded full points if your submissions are not clear.
- The Python notebook must be submitted in the file extension filename.ipynb or filename.py. **Other formats will not be accepted and graded.**
- **Do not zip your files. Zipped files are not considered assignment submissions.**
- You can have one notebook for both questions 2,3(programs).
- After the **Due Date**, late submission of up to one week will incur a 10% loss of total points earned. 5% every week thereafter until the end date.
- Attach this file with self-assessment. This is for your reference if you answered the question completely.